New Uzbekistan University

[CS102]
Computer Programming 1



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ABOUT ME

- Bachelor of Computer Science
  - INHA University in Tashkent (2017-2021)
- M.S. and Ph.D in Artificial Intelligence
  - KAIST, South Korea (2021-2026)
- Visiting Researcher
  - Seoul National University, South Korea (2022-2026)



KAIST AI
Kim Jaechul Graduate School



ABOUT ME

- Assisstant Professor, New Uzbekistan University [2026-]
- Researcher, at KAIST & SNU [2021-2026]
- Senior Software Engineer
  - Multicard (Rahmat) [2021-2025]
  - ATMOS [2019-2021]
  - Turon Telecom (iTV, Cinerama) [2018-2019]



ABOUT ME

- Email: s.kholmatov@newuu.uz
- Office Hours: Tue: 11:00 AM ~ 12:30 PM
- Office: UCA 208

COURSE INTRODUCTION

Subject	Computer Programming 1
ECTS Credits	8
Content	Introduction to Programming in C++ Fundamental data types Numerical systems and binary operations Control Statements Iterators Functions Arrays Pointers I/O Operations

- Textbook & Reference Book
  - *"Introduction to Programming with C++"*, Y. Daniel Liang
  - *"C++ How to Program"*, Harvey Dietel and Paul Dietel

COURSE INTRODUCTION

- Grading Criteria

Grading criteria	Total points
Midterm	30 points
Final	40 points
Quizzes	10 points
Lab Assignments	20 points
Total	100 points

COURSE INTRODUCTION

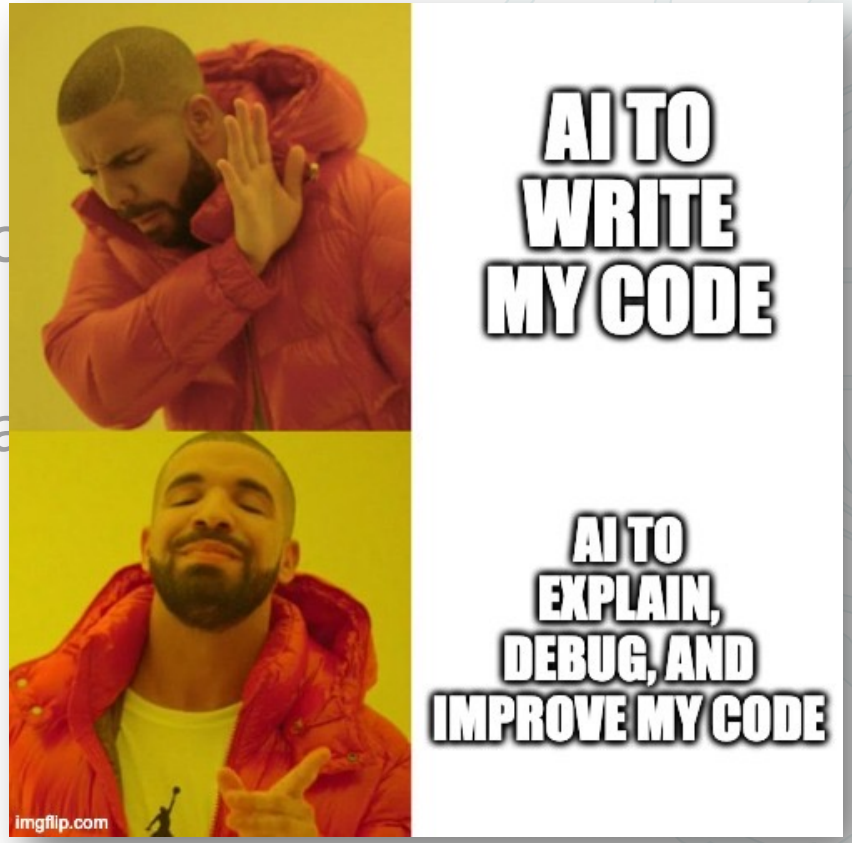
- There are total 5 quizzes in this semester.
  - *Weeks 3, 6, 9, 11 and 13*
- Lab assignments should be submitted during lab hours or during the day of the lab.
- Lab instructors can specify own deadlines for labs. It may vary depending on the schedule.

CLASS POLICIES

- Students are responsible for all missed work, regardless of the reason for absence.
- You can help your peers, but do NOT share your code, otherwise both will get 0 point.
- You can discuss and ask suggestion as much as you want.

CLASS POLICIES

- Students are responsible for all missed work, of the reason for absence.
- You can help your peers, but do NOT share yo otherwise both will get 0 point.
- You can discuss and ask suggestion as much a
- Use AI to **learn**, not to substitute you!



CLASS POLICIES

Missing **25%** of the classes – **FAIL**

OVERVIEW OF PROGRAMMING



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OVERVIEW OF PROGRAMMING

- What is programming?

> Set of instructions that tell a computer what to do

OVERVIEW OF PROGRAMMING

- What are instructions?

Instructions are small commands that tell the CPU what operation to perform.

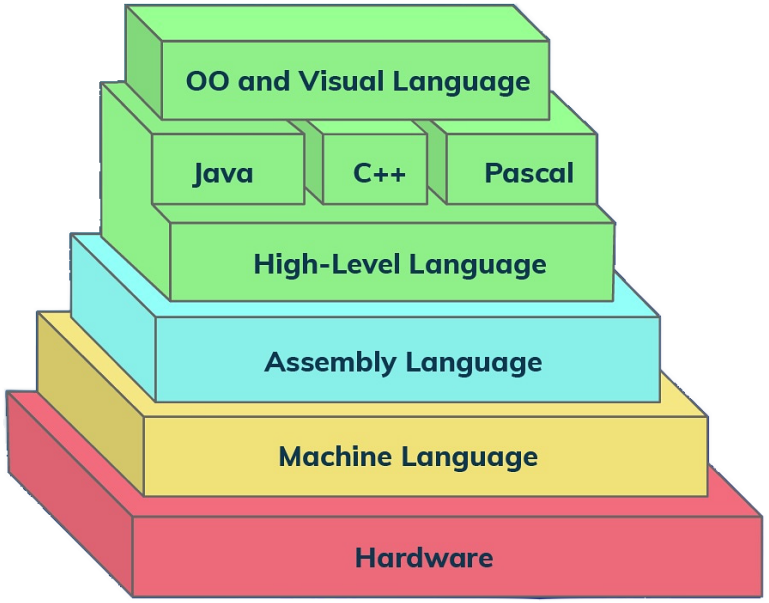
They are encoded as binary data (0s and 1s) according to the CPU's instruction set.
Examples: add/move data or jump to instruction.

OVERVIEW OF PROGRAMMING

- Machine Language

```
100100 010001 ; Load rate
100110 010010 ; Multiply by hours
100010 010011 ; Store the result
```

- Assembly (mnemonics)
  - MOV, MUL, ADD etc.
- High-level languages
  - C++, Java, Python etc.



OVERVIEW OF PROGRAMMING

- Program written in high-level language (e.g., C++) called a *source program* or *source code*

- Source Program must be translated into machine code for execution:
  - **Compiler**
  - **Interpreter**

OVERVIEW OF PROGRAMMING

- Compiler vs. Interpreter

Interpreter (e.g. Python)

Translates and runs the program *line by line* each time it executes.

Compiler (C, C++)

Translates the *whole* program into machine code *once*, producing an executable that runs fast.

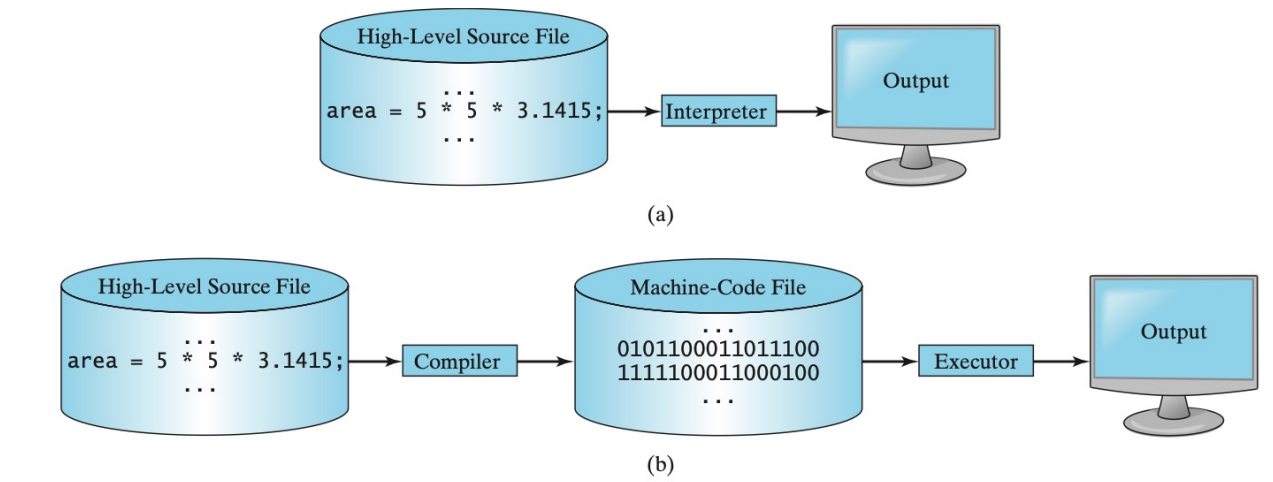


FIGURE 1.9 (a) An interpreter translates and executes a program one statement at a time. (b) A compiler translates the entire source program into a machine-language file for execution.

Number Systems & Data Representations



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Number Systems & Data Representations

- Inside the computer, everything is electrical signals: **on** or **off**.
- Two states are represented with the digits **1** and **0** – *the binary number system*.
- All data – numbers, text, images, sound stored as patterns of 0s and 1s.

Vocabulary

Bit	a single binary digit, 0 or 1
Byte	a group of 8 bits

Number Systems & Data Representations

- Binary -> decimal conversions
  - Example: 1101

$$\underbrace{1}_{\times 2^3}\ \underbrace{1}_{\times 2^2}\ \underbrace{0}_{\times 2^1}\ \underbrace{1}_{\times 2^0} = 8 + 4 + 0 + 1 = 13$$

  - Binary 1101 = decimal 13

Number Systems & Data Representations

- Decimal -> Binary conversions
  - Example: 13, divide by 2 and keep remainder:

$13 \div 2$	$= 6$	remainder **1**
$6 \div 2$	$= 3$	remainder **0**
$3 \div 2$	$= 1$	remainder **1**
$1 \div 2$	$= 0$	remainder **1**

  - Read remainder from bottom-to-top

Number Systems & Data Representations

- Conversions practice
  - Convert binary 11011 to decimal
  - Convert decimal 33 to binary

Number Systems & Data Representations

- Conversions practice
  - Convert binary 11011 to decimal
    - = 2^0 + 2^1 + 2^3 + 2^4 = 1+2+8+16 = 27
  - Convert decimal 33 to binary
    - = 100001

INTRODUCTION TO C++



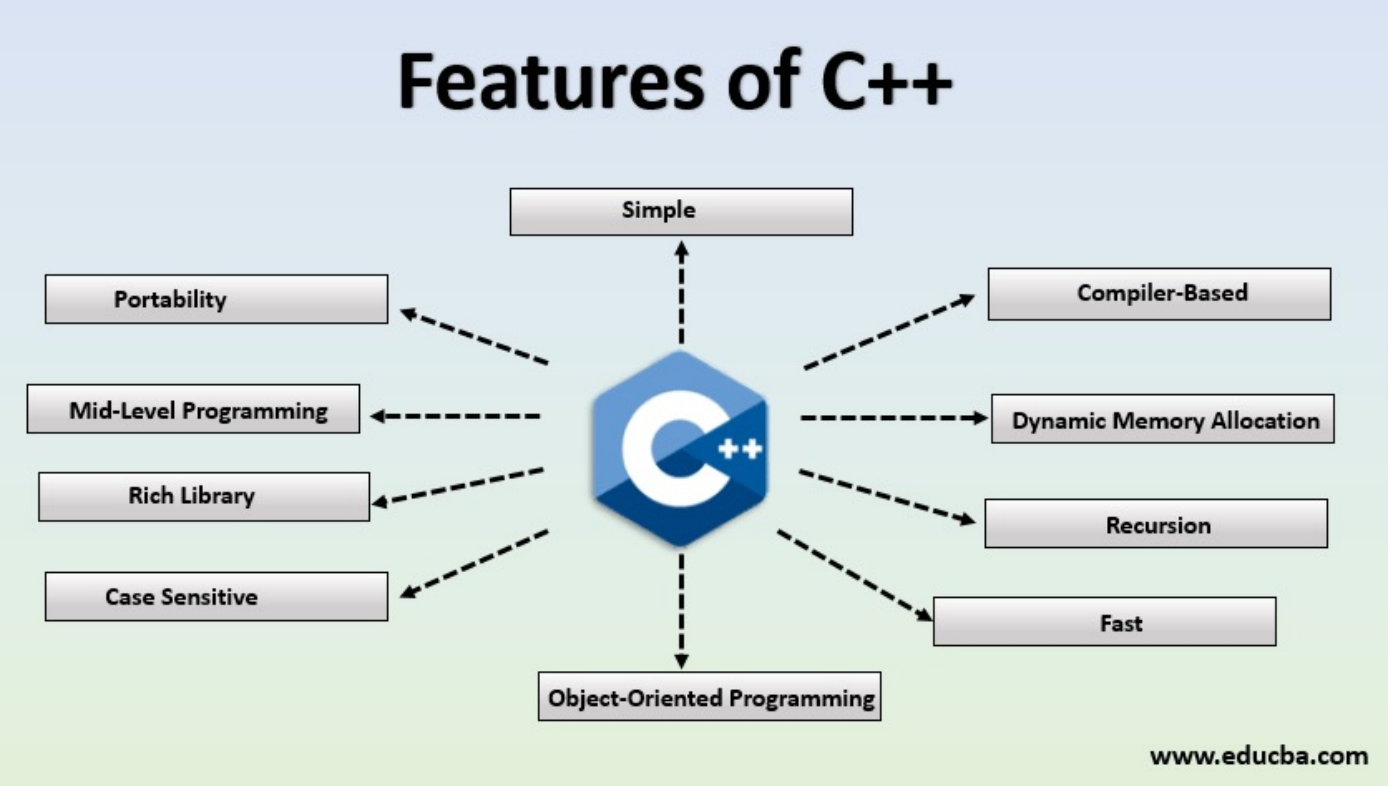
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INTRODUCTION TO C++

- History of C and C++

C	C++
• Evolved from BCPL and B (“typeless” languages). • **Dennis Ritchie**, Bell Labs, 1970s – added data types. • Grew alongside the UNIX operating system.	• **Bjarne Stroustrup**, Bell Labs, early 1980s. • Added object-oriented programming to C. • Programs are built from *classes* and *functions*.

INTRODUCTION TO C++



INTRODUCTION TO C++

- Syntax of C++

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

INTRODUCTION TO C++

- Syntax of C++
  - main() is the entry point

```cpp
#include <iostream> // 1. bring in input/output features
using namespace std; // 2. so we can write cout, not std::cout

int main() { // 3. execution always starts here
    cout << "Hello, World!" << endl; // 4. print + newline
    return 0; // 5. tell the OS: success
}
```

INTRODUCTION TO C++

- Common Errors:
  - Forgetting semicolon ;
  - Mismatching closing/opening braces {} or paratheses ()
  - Misspelling: C++ is case sensitive

```cpp
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using namespace std; // 2. so we can write cout, not std::cout

int main() { // 3. execution always starts here
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Version Control (Git)



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Version Control (Git)

- https://www.youtube.com/watch?v=pBy1zgt0XPc



Telegram Channel

@CS102_Fall_2026

https://t.me/CS102_Fall_2026

